

```
#!pip install streamlit streamlit-drawable-canvas pillow google-generativeai
```

```
import os
```

```
import time
```

```
import base64
```

```
import json
```

```
import re
```

```
import streamlit as st
```

```
from PIL import Image
```

```
from io import BytesIO
```

```
from streamlit_drawable_canvas import st_canvas
```

2026-02-10 22:18:58.819 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode

```
import google.generativeai as genai
```

/usr/local/lib/python3.12/dist-packages/google/colab/_import_hooks/_hook_injector.py:55: FutureWarning:

All support for the `google.generativeai` package has ended. It will no longer be receiving updates or bug fixes. Please switch to the `google.genai` package as soon as possible. See README for more details:

<https://github.com/google-gemini/deprecated-generative-ai-python/blob/main/README.md>

```
loader.exec_module(module)
```

```
from getpass import getpass
```

```
Google_API_KEY = getpass("Enter your Google API key here :")
```

```
Enter your Google API key here :.....
```

Configure Gemini

```
import google.generativeai as genai
```

```
genai.configure(api_key=Google_API_KEY)
```

```
VISION_MODEL = "gemini-pro-vision"
```

```
TEXT_MODEL = "gemini-pro"
```

```
SYNONYMS = {
    "car": ["automobile", "vehicle"],
    "cat": ["kitty", "kitten"],
    "dog": ["puppy"],
    "cup": ["mug"],
    "house": ["home"],
    "tree": ["plant"],
    "phone": ["mobile"],
}
```

Word Generator

```
def generate_words():
    model = genai.GenerativeModel(TEXT_MODEL)

    prompt = """
    Generate 10 very simple, single-word noun objects that are easy to draw in under 20 seconds.

    Rules:
    - Only common physical objects
    - No abstract concepts
    - No multi-word phrases
```

```

- Output ONLY valid JSON
- JSON format: {"words": ["cat", "house", ...]}
"""

response = model.generate_content(
    prompt,
    generation_config={"temperature": 0.6}
)

try:
    data = json.loads(response.text)
    return data["words"]
except json.decoder.JSONDecodeError:
    st.warning("Model did not return valid JSON for word generation. Please try again.")
    return ["cat", "dog", "house", "tree", "car", "cup", "phone", "book", "hat", "star"]

```

Drawing Canvas

```

def draw_canvas():
    canvas = st_canvas(
        fill_color="rgba(255, 255, 255, 0)",
        stroke_width=5,
        stroke_color="#000000",
        background_color="#FFFFFF",
        height=300,
        width=300,
        drawing_mode="freedraw",
        key="canvas"
    )
    return canvas

```

Convert Canvas to Base64

```

def image_to_base64(image):
    buffer = BytesIO()
    image.save(buffer, format="PNG")
    return base64.b64encoder(buffer.getvalue()).decode()

```

Evaluation Logic

```

def normalize(text):
    text = text.lower()
    text = re.sub(r"\b(a|an|the)\b", "", text)
    return text.strip()

```

```

def evaluate(guess, target):
    return normalize(guess) == normalize(target)

```

Gemini Guess

```

def gemini_guess(image_base64):
    model = genai.GenerativeModel(VISION_MODEL)

    prompt = """
    You are playing a drawing guessing game.

    Rules:
    - Guess ONLY ONE Word.
    - No Explanations
    - If unsure, return "Uncertain".
    - Output ONLY valid JSON.

    JSON format:
    {"guess": "<one_word_here>"}
    """

    image_part = {
        "mime_type": "image/png",
        "data": base64.b64decode(image_base64)
    }

    response = model.generate_content(
        [prompt, image_part],
        generation_config={"temperature": 0.2}
    )

```

```

try:
    return json.loads(response.text)["guess"]
except json.decoder.JSONDecodeError:
    st.warning("Model did not return valid JSON for guessing. Returning 'Uncertain'.")
    return "Uncertain"

```

Streamlit Orchestration

```

st.title("🐼 Gemini QuickDraw Clone")

if "words" not in st.session_state:
    try:
        st.session_state.words = generate_words()
        st.session_state.target = st.session_state.words[0]
    except Exception as e:
        st.error(f"Failed to generate words: {e}")
        # Ensure target is still set to prevent AttributeError, using a default fallback
        st.session_state.words = ["default"] # Assign a default list to words
        st.session_state.target = st.session_state.words[0] # Assign target from the default list
        st.stop() # Attempt to stop further execution in this run

st.write(f"**Draw this:** `{st.session_state.target}`")

canvas = draw_canvas()

if st.button("Submit Drawing"):
    if canvas.image_data is not None:
        start = time.time()

        image = Image.fromarray(canvas.image_data.astype("uint8"))
        img_b64 = image_to_base64(image)

        guess = gemini_guess(img_b64)
        duration = round(time.time() - start, 2)

        correct = evaluate(guess, st.session_state.target)

        st.subheader(f"🐼 Gemini guessed: **{guess}**")
        st.write(f"🕒 Time taken: {duration}s")

        if correct:
            st.success("✅ Correct!")
        else:
            st.error("❌ Incorrect")

```

2026-02-10 22:19:59.905 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode
 2026-02-10 22:19:59.977

Warning: to view this Streamlit app on a browser, run it with the following command:

```

streamlit run /usr/local/lib/python3.12/dist-packages/colab_kernel_launcher.py [ARGUMENTS]
2026-02-10 22:19:59.977 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode
2026-02-10 22:19:59.979 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode
2026-02-10 22:19:59.979 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode
2026-02-10 22:19:59.980 Session state does not function when running a script without `streamlit run`
2026-02-10 22:20:01.072 404 POST /v1beta/models/gemini-pro:generateContent?%24alt=json%3Benum-encoding%3Dint (:::1) 1083.13ms
2026-02-10 22:20:01.073 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode
2026-02-10 22:20:01.074 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode
2026-02-10 22:20:01.074 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode
2026-02-10 22:20:01.075 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode
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2026-02-10 22:20:01.077 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode
2026-02-10 22:20:01.078 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode
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2026-02-10 22:20:01.081 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode
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2026-02-10 22:20:01.083 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode
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2026-02-10 22:20:01.085 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode
2026-02-10 22:20:01.577 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode
2026-02-10 22:20:01.578 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode
2026-02-10 22:20:01.579 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode
2026-02-10 22:20:01.579 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode
2026-02-10 22:20:01.580 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode
2026-02-10 22:20:01.582 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode
2026-02-10 22:20:01.583 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode
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```

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